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Citizen science generates reliable migration chronologies to supplement professional surveys for waterfowl habitat management

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ABSTRACT

Targeted monitoring by trained professionals has been the standard to inform evidence-based regulatory and conservation planning decisions for migratory waterfowl and other wildlife in North America. As inferential expectations derived from professional surveys grow and funding wanes, decision-makers are turning toward citizen science as supplemental or alternative data sources. However, external validation of citizen science data, in cooperation with experts, is recommended before their use in management decisions. We compiled data from professional surveys across 7 U.S. states within the Upper Mississippi / Great Lakes Joint Venture (JV) for 16 species of waterfowl (ducks, geese, and swans) to compare with and potentially validate temporal trends in weekly relative abundance predictions (i.e., migration curves) by eBird citizen science data. We demonstrated that eBird weekly relative abundance produced similar migration curves to those derived from professional surveys for most species. Concordance between migration curves for uncommon or vagrant species was less reliable, especially at the county level. However, at spatial scales more relevant for regulatory or conservation planning decisions (e.g., state or JV scale), correlation between eBird and professionally derived migration curves increased to adequate levels of concordance for most species ($\rho \geq 0.70$). We concluded that eBird weekly relative abundance is a suitable supplement to professional surveys to generate migration curves for waterfowl conservation planning during the nonbreeding season. Additionally, states and JVs that lack monitoring at appropriate temporal frequency may consider using eBird relative abundance to derive migration chronologies for priority species that are well-distributed throughout their region. Broadly, we encourage continued external validation of citizen science data—which requires close partnership between researchers and decision-makers—for its wise use in management decisions.

Keywords: Anatidae, conservation planning, community science, ducks, eBird, nonbreeding, North American Waterfowl Management Plan, strategic habitat conservation, monitoring, waterfowl

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LAY SUMMARY

- Understanding when and where birds occur is critical for effective conservation planning for migratory waterfowl, yet funding for monitoring programs by wildlife professionals continues to decline.
- We evaluated whether migration data derived from eBird relative abundance supplement professionally collected waterfowl count data by comparing these 2 data sources at 3 spatial scales relevant to management across 7 U.S. states in the Upper Mississippi / Great Lakes Joint Venture region.
- Migration chronology based on eBird relative abundance data corresponded well with professionally collected count data, particularly for common and widely distributed species upon which most waterfowl conservation planning decisions are based.
- Results suggest that eBird relative abundance could be used as complementary data to understand the timing and distribution of waterfowl migration to inform habitat management and conservation for waterfowl during autumn–winter, especially where professional monitoring is limited or absent.

La ciencia ciudadana genera cronologías migratorias confiables para complementar los relevamientos profesionales en la gestión del hábitat de aves acuáticas

RESUMEN

El monitoreo realizado por profesionales capacitados ha sido el estándar para orientar las decisiones regulatorias y de planificación de conservación basadas en evidencia para las aves acuáticas migratorias y otras especies silvestres en América del Norte. A medida que aumentan las expectativas inferenciales derivadas de los relevamientos profesionales y disminuyen los fondos, los responsables de la toma de decisiones están recurriendo a la ciencia ciudadana como fuente de datos complementaria o alternativa. Sin embargo, se recomienda la validación externa de los datos de ciencia ciudadana, en cooperación con expertos, antes de su uso en decisiones de manejo. Reunimos datos de relevamientos profesionales en 7 estados de EUA en el marco de la Iniciativa Conjunta (IC) del Alto Misisipi / Grandes Lagos para 16 especies de aves acuáticas (patos, gansos y cisnes), para comparar y potencialmente validar las tendencias temporales en las predicciones semanales de abundancia relativa (i.e., curvas migratorias) a partir de datos de ciencia ciudadana de eBird. Demostramos que la abundancia relativa semanal de eBird produjo curvas migratorias similares a las derivadas de los relevamientos profesionales para la mayoría de las especies. La concordancia entre las curvas migratorias para especies poco comunes o errantes fue menos confiable, especialmente a nivel de condado. Sin embargo, a escalas espaciales más relevantes para las decisiones regulatorias o de planificación de conservación (e.g., a nivel estatal o de IC), la correlación entre las curvas migratorias derivadas de eBird y las profesionales aumentó a niveles adecuados de concordancia para la mayoría de las especies ($\rho \geq 0.70$). Concluimos que la abundancia relativa semanal de eBird es un complemento adecuado para los relevamientos profesionales para generar curvas migratorias para la planificación de la conservación de aves acuáticas durante la temporada no reproductiva. Además, los estados y las IC que carecen de monitoreo con la frecuencia temporal adecuada pueden considerar usar la abundancia relativa de eBird para derivar cronologías migratorias de especies prioritarias que están bien distribuidas en su región. En términos generales, alentamos la continua validación externa de los datos de ciencia ciudadana—la cual requiere una estrecha colaboración entre investigadores y tomadores de decisiones—para su uso responsable en decisiones de manejo.

Palabras clave: Anatidae, aves acuáticas, ciencia comunitaria, conservación estratégica del hábitat, eBird, monitoreo, no reproductivo, patos, Plan de Manejo de Aves Acuáticas de América del Norte, planificación de conservación

INTRODUCTION

Resource requirements of migratory birds vary across the annual cycle, so understanding seasonal movements and settling patterns is necessary to identify important migratory and wintering habitats (Newton 2008, Marra et al. 2015). Detailed information on bird abundance and migration chronology is needed to prioritize areas for protection or restoration (Reynolds et al. 2017, Masto et al. 2023, 2024a), ensure sufficient resources are available at appropriate times (O'Neal et al. 2012, Reynolds et al. 2017, Donnelly et al. 2019), and mitigate threats to migratory populations, including habitat loss and resource constraints imposed by climate change (Robinson et al., 2009, Both et al. 2010, Runge et al. 2014). Thus, key metrics needed for effective conservation of migratory birds include species-specific abundance, stopover duration, and changes in abundance in space and time (i.e., bird turnover; Soulliere et al. 2013, Runge et al. 2014, Stillman et al. 2023, Nussbaumer et al. 2024).

Conservation and management decisions for migratory birds continue to be informed by targeted population monitoring efforts conducted by trained wildlife professionals (Nichols and Williams 2006, Wintle et al. 2010). For example, the U.S. Fish and Wildlife Service (USFWS) operationalized the Waterfowl Breeding Population and Habitat Survey in 1955 to monitor waterfowl breeding abundance and wetland conditions, which continues to inform population management and regulatory processes (Smith 1995, USFWS 2023). However, financial constraints and concerns for observer safety, combined with demand for increased scientific rigor in data collections and underlying environmental changes, often hamper the consistency with which agencies can conduct monitoring efforts and/or the usefulness of the data collected (Eggeman and Johnson 1989, Kéry and Schmidt 2008, Soulliere et al. 2013).

Unlike targeted monitoring efforts, crowdsourced citizen science initiatives are often expansive in geographical and taxonomic scope, while being relatively inexpensive to maintain (Sullivan et al. 2014, Kelling et al. 2019). Recently, high-quality citizen science data have been used to extend (Stillman et al. 2023), reallocate (Stuber et al. 2022), or supplement targeted monitoring efforts (Robinson et al. 2020, Howell et al. 2022; Table 1 in Stuber et al. 2022). Management agencies and conservation planners are increasingly turning to high-quality citizen science data to inform decision-making, especially when targeted monitoring efforts are insufficient (e.g., Soulliere et al. 2017, Lancaster et al. 2021, Hagy et al. 2022, Stillman et al. 2023). However, it is important that citizen science datasets are used appropriately, in direct partnership with decision-makers, and that crowdsourced data are validated internally and externally to verify their utility (Baker et al. 2021, Ruiz-Gutierrez et al. 2021).

Waterfowl management in North America is a premier example of successful wildlife conservation (Williams and

Castelli 2012, Anderson and Padding 2017), in part because the North American Waterfowl Management Plan (NAWMP) provides a framework connecting continental population objectives, intensive monitoring efforts, and habitat conservation at regional scales across the full annual life cycle of ducks, geese, and swans (U.S. Department of Interior 1986 and NAWMP revisions). The NAWMP encouraged partnerships among state, provincial, territorial, federal, and non-governmental agencies to monitor and conserve waterfowl within their respective geographical regions (hereafter, Migratory Bird Joint Ventures [JVs]); by stepping-down continental population objectives to regional scales, habitat delivery for migratory waterfowl became more tractable for the public and private partnerships within those geographies. Conservation planning and habitat priorities for waterfowl during the nonbreeding season assume that dietary energy from food resources is the factor most likely to limit populations during autumn and winter (i.e., food limitation hypothesis; Sutherland et al. 1998, Danner et al. 2013, Williams et al. 2014). Consequently, JVs use bioenergetic models to translate regional waterfowl population objectives into estimates of energy demand (kcal), which, in turn, can be converted into habitat objectives based on habitat-specific estimates of food type, quantity, and caloric values (i.e., energy supply). Thus, data on population size, proportional distributions among JVs, and species-specific migration chronologies (i.e., relative abundance over time) are required to estimate time-specific energy needs for nonbreeding waterfowl (Soulliere et al. 2013, Williams et al. 2014).

Migration chronology is an important parameter because bioenergetic models are especially sensitive to abundance over time in a given region (i.e., the shape of the migration curve) when determining period-specific energy demand (Ringelman et al. 2018, Lancaster et al. 2021). Species-specific migration chronologies are often derived from local or regional population surveys conducted by wildlife professionals. However, these professional surveys are rarely coordinated within or among JVs (Soulliere et al. 2013). Additionally, they have been criticized for a lack of scientific rigor (Eggeman and Johnson 1989) and face increased financial constraints that reduce the likelihood of continuation into the future (Masto et al. 2021). In a few cases, waterfowl practitioners have used Cornell Lab of Ornithology's raw eBird data (Sullivan et al. 2014) or the modeled eBird Status and Trends Products (Fink et al. 2020) to derive migration chronology for local or regional conservation planning because professionally collected data were deemed inadequate (Soulliere et al. 2013, 2017; Lancaster et al. 2021, Hagy et al. 2022). Importantly, however, eBird-derived migration chronologies were not externally validated against trusted data sources, which could bias estimated energy allocation efforts in these and other use cases (sensu Ruiz-Gutierrez et al. 2021). Given the increasing interest in and use of these crowdsourced data products among the waterfowl management community, external validations are necessary to test critical assumptions and ensure the proper use of eBird products in bioenergetic modeling for waterfowl conservation planning and habitat delivery.

We tested the validity of using eBird to generate migration chronologies for regional conservation planning by comparing eBird-derived migration curves to those generated by professionally conducted surveys across 7 U.S. states within the Upper Mississippi / Great Lakes JV region. Specifically, we compared migration chronologies of 16 waterfowl species between the 2 data streams at 3 spatial scales: (1) the county or multi-county; (2) the U.S. state; and (3) JV region. We predicted a priori that migration chronologies would align well, with increasing concordance and overlap as the spatial scale of comparisons increased. Additionally, we hypothesized that a species' occurrence (≥ 1 bird) or abundance ($1 \ldots n$ birds) may affect concordance of migration curves derived from eBird compared to those generated from professional surveys due to different sampling procedures. Specifically, we predicted lower concordance of migration chronologies for species observed less frequently or at low densities and vice versa. We expect results from our study to provide evidence for or against the use of eBird-derived migration chronologies within conservation planning frameworks for North American waterfowl during the nonbreeding season.

METHODS

Targeted Monitoring

State and federal conservation agencies and researchers use several survey types to monitor nonbreeding waterfowl population abundance and distribution (Soulliere et al. 2013, Stuber et al. 2022). For example, some implement design-based aerial transect surveys to estimate waterfowl abundance and distributions (e.g., Pearse et al. 2008, Masto et al. 2021), others use aerial cruise surveys (e.g., Illinois, Ohio; Hennig et al. 2017, Gilbert et al. 2021), while others conduct ground and water-based inventories from a vehicle across entire properties (e.g., National Wildlife Refuges [NWRs]; Iowa, Indiana, Kansas, Michigan, Missouri; Soulliere et al. 2013, Loges et al. 2021). Regardless of survey type, targeted monitoring efforts are typically conducted by wildlife professionals. These surveys are internally vetted for reliability and consistency, and are assumed to reasonably index waterfowl abundance, distributions, and migration timing. For our study, we used count data from targeted monitoring efforts conducted by state and federal conservation agencies and University partners. These surveys served as our professionally collected survey data from which we derived migration curves (i.e., counts through time within a specified geographic region). Specifically, we used count data from autumn through winter 2022–2023 aerial cruise surveys conducted by the Illinois Natural History Survey, Illinois Department of Natural Resources, and Ohio Department of Natural Resources. Surveys in Illinois were conducted at weekly intervals, while surveys in Ohio were conducted at weekly intervals early in the season (September) and twice per month during November and December. We also used data from ground-based vehicular surveys conducted by Kansas Department of Wildlife and Parks, Missouri Department of Conservation, and Indiana, Iowa, and Michigan Departments of Natural Resources during the same time period. Surveys in Kansas were conducted twice per month; all other ground-based surveys were conducted weekly (Figure 1).

eBird Relative Abundance

eBird is a citizen science initiative that uses semi-structured data collection methods by hundreds of thousands of participants annually to monitor bird occurrence, abundance, and distributions globally (Sullivan et al. 2014, Kelling et al. 2019). Following internal review (Lagoze 2014, Baker et al. 2021), the Cornell Lab of Ornithology's eBird Status and Trends Team filters raw eBird data to account for spatial, observer,

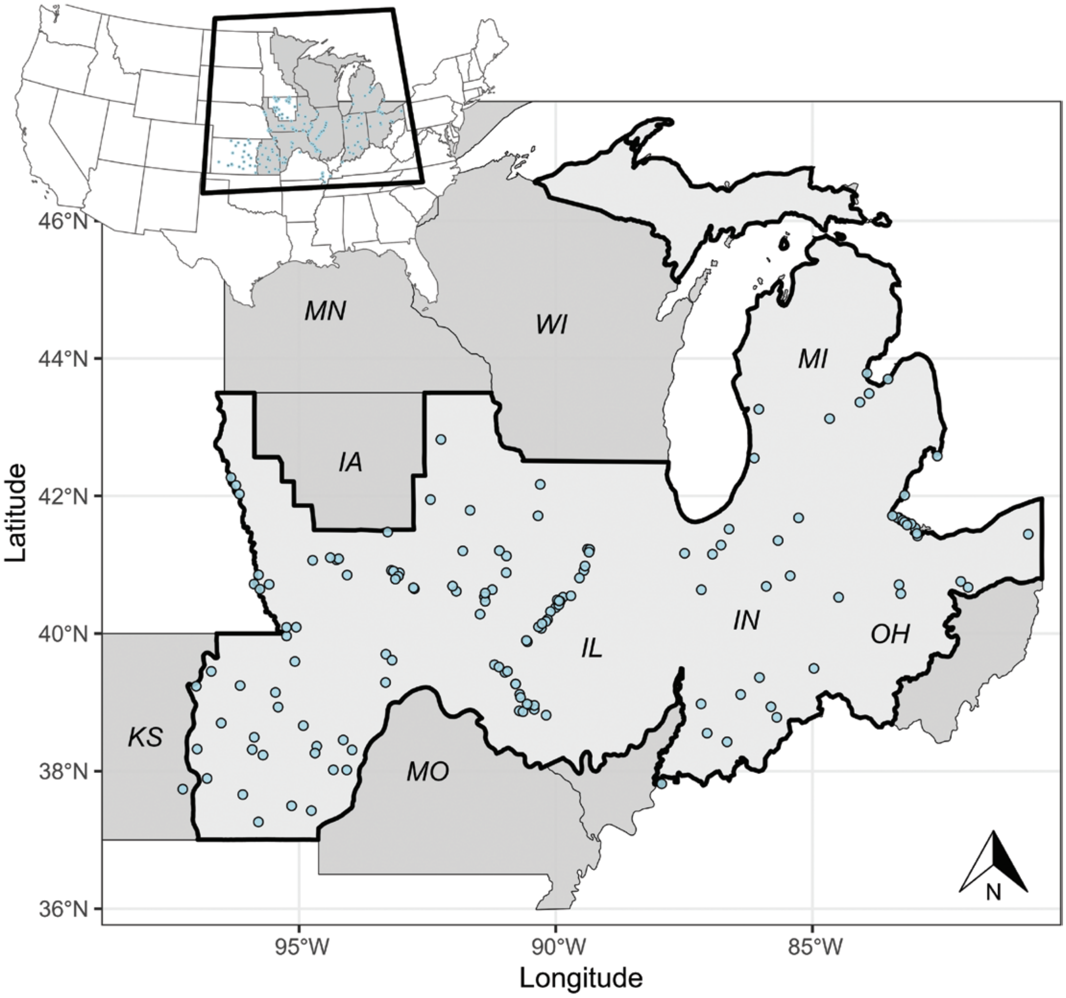

FIGURE 1. Study area encompassed the Upper Mississippi / Great Lakes Joint Venture region (JV; gray in the inset map). Professionally collected survey data (blue in the inset map) spanned 7 U.S. states including Kansas (KS), Iowa (IA), Missouri (MO), Illinois (IL), Indiana (IN), Michigan (MI), and Ohio (OH). When analyzed at the Joint Venture scale, the JV boundary was modified to remove portions of Minnesota and Wisconsin where professionally collected survey data was absent (light gray and bold outline) and professionally collected survey data was clipped to this modified boundary (blue locations).

and effort-based biases (Johnston et al. 2021). Specifically, the eBird Status Product (hereafter, eBird relative abundance) is limited to: (1) checklists between 2008 to the present; (2) only "complete" checklists; and (3) checklists with stationary protocols or travel distances < 5 km (Fink et al. 2023a). Additionally, duplicate checklists, those with > 10 observers, or those that took longer than 5 hr are removed (Fink et al. 2023a). The resulting 53 million checklists are annotated with 147 spatially and temporally explicit variables that capture landscape associations (e.g., landcover, surface water, bathymetry) and account for the data generation and detection processes (e.g., coordinates, weather, observer skill, and search-effort; Robinson et al. 2017; Johnston et al. 2018, 2021).

Highly filtered and annotated checklists were then used to train statistical and machine learning algorithms that generate predictions of species' occurrence, relative abundance, distributions, and population trends (see Fink et al. 2020, 2023a, 2023b for further description). Specifically, the eBird Status Product uses the relationships between bird occurrence and abundance, observer and effort-based variables, and environmental descriptors from model outputs to predict species-specific weekly relative abundance at a ~3 × 3 km spatial resolution by setting predictor variables to their weekly or yearly averages in time and space. Therefore, we used the eBird Status Product to generate eBird-derived waterfowl migration curves predicted to 2022–2023 (i.e., prediction through time within a specified geographic region). The eBird Status Product (hereafter, eBird relative abundance) is defined as the "expected count of a given species observed by an expert eBirder at the optimal time of day while expending necessary effort to maximize species detection" (Fink et al. 2023a). In summary, modeled eBird relative abundance predictions include > 10 yr of highly filtered eBird checklists, 147 environmental, landcover, and detection covariates, and machine learning models that predict a species relative abundance and distributions in space and time (Fink et al. 2020). Therefore, we expected eBird relative abundance to align with professionally conducted survey data both spatially and temporally, and therefore generate similar species-specific migration curves.

Geoprocessing and Spatiotemporal Alignment

Our primary goal was to evaluate the degree of concordance between migration chronologies derived from eBird relative abundance and professionally collected survey data across 3 spatial scales relevant for local or regional conservation planning (county or multi-county < U.S. state < Joint Venture). We began at county or multi-county scales because many bioenergetic models use a method of stepping-down continental waterfowl population objectives to the county level using harvest records to describe duck distributions and subsequent energy demand within a region (Fleming et al. 2017, 2019). Furthermore, waterfowl abundance and turnover at such fine scales are useful in providing detailed insight into resource allocation, habitat management, and restoration (e.g., NWRs; Ringelman et al. 2018, Hagy et al. 2022). To describe temporal variation in waterfowl abundance, we summed species-specific count data from professional surveys to the county level. In cases where area count data intersected 2 or more counties, we combined counties and then summed the species-specific count data to this multi-county region.

Waterfowl abundance and migration chronologies at the state-level are often used to inform waterfowl hunters and birders of the abundance and distribution of ducks and geese, as well as migration timing across time and space during autumn and winter arrival (Pearse et al. 2008, Forbes Biological Station 2020, Masto et al. 2021). State-level migration chronologies are also useful for regulatory processes, such as refining waterfowl hunting season dates, zones, or evaluating habitat restoration initiatives (Soulliere et al. 2013, Stiller et al. 2022). We followed the same procedure described above for the state-level analyses, summing weekly species-specific count data from professionally conducted surveys across each U.S. state (n = 7). We assumed, as do state conservation agencies, that aerial or ground-based surveys adequately represented duck distribution and turnover for the entire state (Figure 1).

The largest spatial scale in our analyses was the JV-region scale, specifically using the boundary of the Upper Mississippi / Great Lakes JV region. Although coarse, the administrative boundaries of JVs are typically the scale at which bird abundance objectives and migration chronology are combined to estimate waterfowl energy demand, and in turn, regional habitat objectives are established to support target populations (Williams et al. 2014, Fleming et al. 2017, 2019). Two geospatial data management procedures were required at the JV scale: (1) we excluded professional surveys in Iowa, Kansas, and Missouri that were located outside of the Upper Mississippi / Great Lakes JV boundary; and (2) we modified the JV boundary by removing the portions of Minnesota and Wisconsin, where professionally collected survey data during the nonbreeding season were absent (Figure 1). These 2 procedures ensured spatial alignment of professional surveys and eBird relative abundance at the JV scale. We then summed weekly species-specific counts across professionally collected survey data within the modified JV boundary to represent migration chronology at the largest spatial scale (Figure 1C). We assumed weekly summation of professionally collected count data used to generate JV-level migration curves were unaffected by differences in survey effort or methodology among U.S. states.

We used the *ebirdst* package (Strimas-Mackey et al. 2023) in R version 4.3.2 (R Core Team 2024) to extract eBird relative abundance for each species within the specified geographies described above. We summed weekly eBird relative abundance predictions across the number of ~3 × 3 km pixels for each species and within each specified geography to align spatially with our professionally collected survey data. For each data stream, we created a vector of weeks within the year to temporally align professionally collected data and eBird relative abundance.

Validation

For validation analyses, we scaled professionally collected count data and eBird relative abundance predictions to a proportion of their maximum estimates, where 1 indicated the highest count or prediction in the time-series, and all other weekly abundance estimates fell between 0 and 0.99 (Soulliere et al. 2017, Stuber et al. 2022). This transformation placed each data set on the same scale, enabling direct comparison between them. We used a Spearman's rank correlation analysis (ρ) to compare temporal trends in estimated waterfowl abundance between eBird relative abundance and professional surveys during 2022–2023 (Stuber et al. 2022). Spearman's correlation evaluated the rank for each species-specific weekly abundance estimate for each data set (i.e., highest, 2nd highest, lowest) and therefore, ρ represented the overall concordance between the 2 migration-chronology time-series that was robust to counting or prediction error for each spatial scale and species combination. We also computed the area of overlap (∩) between migration curves derived from professional surveys and eBird relative abundance for each spatial scale and species combination using the *overlapping* package in R (Pastore 2018). Specifically, we estimated the kernel density of migration curves to smooth the curves to approximate an empirical probability density function for each data set. We then used numerical integration to estimate the percent overlapping area under the smoothed curves based on the integral of the minimum between the 2 migration curves (Pastore and Calcagnì 2019). Overlapping area under the smoothed migration curve provided an additional measure of concordance that was more robust to sparse data and abundant zeros compared to ρ.

Effects of Abundance on Concordance

We suspected species' abundance would influence concordance between the 2 data sets. In particular, we predicted migration chronologies would be less correlated for species observed at lower densities and vice versa. Therefore, we modeled variation in Spearman's correlation coefficient (ρ) at the county or multi-county scale relative to the raw maximum species-specific estimate of abundance from professional surveys. Because ρ is a naturally bounded statistic between −1 and 1, we modeled variation in ρ with a beta distribution *Beta* (μ, ϕ) (Douma and Weedon 2019). We standardized $\frac{\rho+1}{2}$ to conform to ordinary beta regression where the response variable is bounded between 0 and 1. The precision parameter ϕ was estimated as a single value for each species-specific model to avoid overfitting. Model structure was:

$$\rho_{i,\ sp} \sim (\beta_0 + \alpha_{\text{State}}) + \beta_{\text{maximum\_abundance}\ i,\ sp} + \varepsilon_{i,sp}$$

where correlation of migration chronologies between eBird relative abundance and professional surveys (ρ) for observation i and species sp was a function of the maximum recorded relative abundance of the species in professional surveys (β). We specified random intercepts for each U.S. state (α_{State}) to account for spatial correlation of county or multi-counties nested within U.S. state jurisdictional boundaries (5–7 U.S. states depending on species), and a residual error term $\varepsilon_{i,sp}$ for each species-specific model (i.e., 16 species). We fitted models in the *brms* package in R (Bürkner 2017). We specified

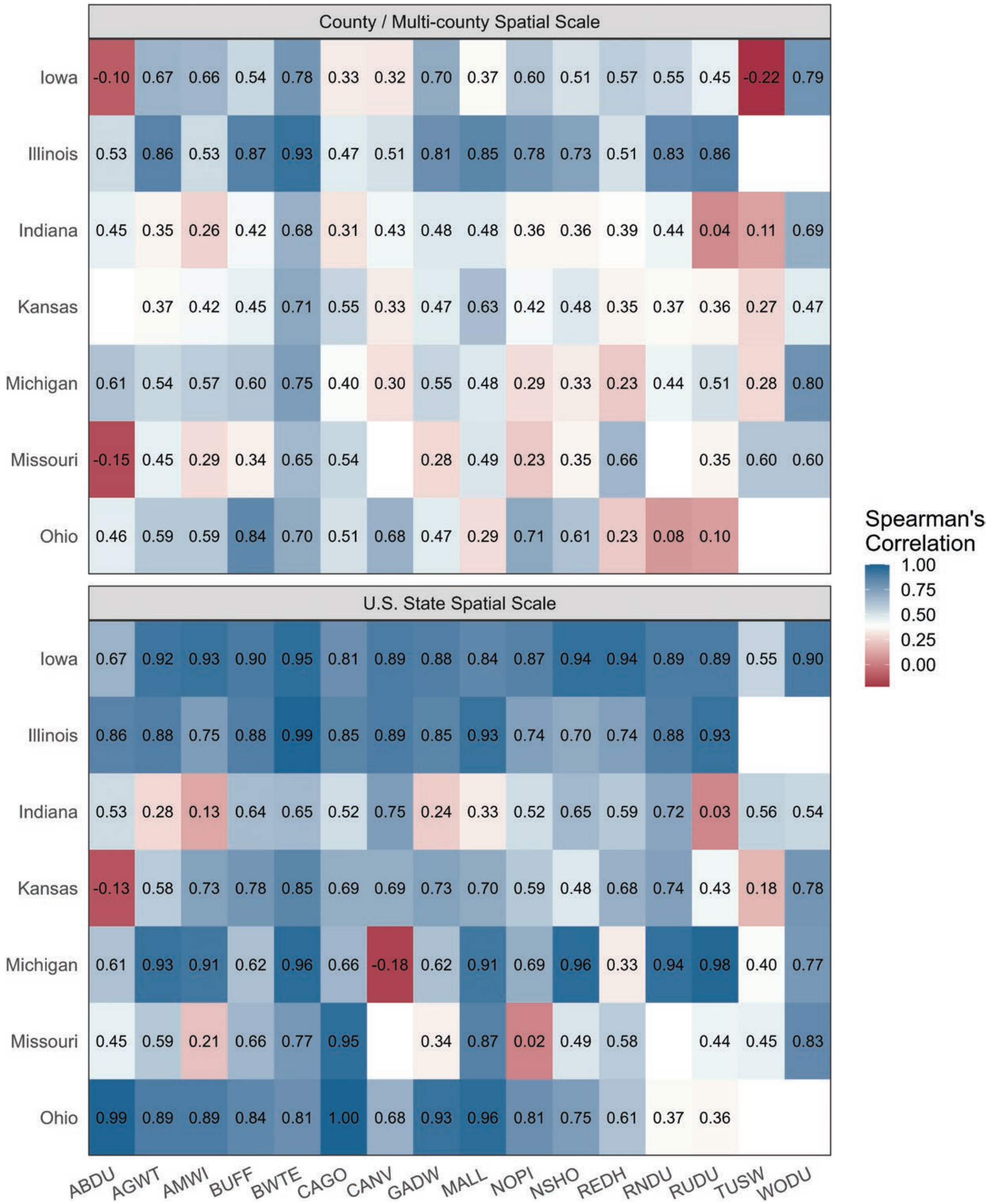

FIGURE 2. Mean Spearman's correlation between migration curves derived from professional surveys and eBird relative abundance for each state when averaged across county / multi-counties for 16 species of waterfowl (top panel) and Spearman's correlations when comparisons were made at the state-level (bottom panel). Color ramps are the same for each plot based on the range and midpoint across both data sets; therefore, the plot is intended to illustrate the variability in concordance at the county / multi-county scale (top panel) and the improvement or change in concordance ($\Delta\rho$) when migration curves were compared at an increased spatial scale more relevant for regulatory or planning purposes (bottom panel). See Supplementary Material Tables S1 and S2 for standard errors for mean concordance values for counties / multi-counties, percent overlap across spatial scales, and $\Delta\rho$ between county / multi-county and state-level spatial scales. Species included were *Anas rubripes* (ABDU), *Anas carolinensis* (AGWT), *Mareca americana* (AMWI), *Bucephala albeola* (BUFF), *Spatula discors* (BWTE), *Branta canadensis* (CAGO), *Aythya valisineria* (CANV), *Mareca strepera* (GADW), *Anas platyrhynchos* (MALL), *Anas acuta* (NOPI), *Spatula clypeata* (NSHO), *Aythya americana* (REDH), *Aythya collaris* (RNDU), *Oxyura jamaicensis* (RUDU), *Cynus colombianus* (TUSW), and *Aix sponsa* (WODU).

weakly informative normal priors ($\mu = 0$ and $\sigma = 1$) and used 3 Markov Monte Carlo chains with 10,000 iterations, 4,000 burn-in iterations, and a thinning rate of 1. We confirmed convergence when $\hat{R} \leq 1.10$ and by visually examining traceplots (Brooks and Gelman 2012). We present posterior medians and 90% credible intervals (CRI) as measures of centrality and variance, respectively. We interpreted the effect size as biologically meaningful when CRIs did not overlap zero. We also report back-transformed model predictions and associated 90% CRIs graphically.

RESULTS

We compared migration curves derived from weekly or biweekly professional surveys to those derived from weekly predictions from eBird relative abundance at 3 spatial scales (i.e., county or multi-county, U.S. state, and JV region) for 16 species of waterfowl: *Anas rubripes* (American Black Duck), *A. carolinensis* (American green-winged Teal), *Mareca americana* (American Wigeon), *Bucephala albeola* (Bufflehead), *Spatula discors* (Blue-winged Teal), *Branta canadensis* (Canada Goose), *Aythya valisineria* (Canvasback), *Mareca strepera* (Gadwall), *Anas platyrhynchos* (Mallard), *A. acuta* (Northern Pintail), *S. clypeata* (Northern Shoveler), *Aythya americana* (Redhead), *A. collaris* (Ring-necked Duck), *Oxyura jamaicensis* (Ruddy Duck), *Cynus colombianus* (Tundra Swan), and *Aix sponsa* (Wood Duck).

Scale-dependent Validation

Visual inspection of migration curves between eBird relative abundance and professionally conducted surveys indicated variable concordance between the 2 datasets at the county or multi-county spatial scale (Figure 2A). The Spearman's correlation, when averaged across counties and species, was greatest in Illinois and least in Missouri (Figure 2). Specifically, mean ρ ± standard deviation (SD) was 0.71 ± 0.25 for Illinois ($n = 116$), 0.54 ± 0.32 for Iowa ($n = 400$), 0.45 ± 0.30 ($n = 498$) for Kansas, 0.45 ± 0.41 for Ohio ($n = 53$), 0.44 ± 0.38 for Michigan ($n = 123$), 0.42 ± 0.31 for Indiana ($n = 231$), and 0.39 ± 0.31 for Missouri ($n = 196$; Figure 2A; Supplementary Material Table S1). When migration curves were smoothed at the county or multi-county scale, mean percent overlap between eBird relative abundance and professionally conducted surveys ($A \cap B$) was 0.70 ± 0.20 for Illinois ($n = 116$), 0.69 ± 0.19 for Iowa ($n = 400$), 0.62 ± 0.23 for Kansas ($n = 492$), 0.65 ± 0.23 for Ohio ($n = 53$), 0.65 ± 0.21 for Michigan ($n = 123$), 0.62 ± 0.21 for Indiana ($n = 231$), and 0.59 ± 0.23 for Missouri ($n = 196$; Supplementary Material Table S1).

As the spatial scale of comparisons increased from county or multi-county to the U.S. state level, migration curves were increasingly correlated (Figure 2; Supplementary Material Tables S2 and S3). For example, the difference in mean ρ when averaged across counties or multi-counties compared to ρ at the U.S. state-level ($\Delta\rho$) increased for all 16 species in Iowa ($\Delta\rho$ range: 0.18–0.77), 12 of 14 species in Illinois ($\Delta\rho$ range: −0.04 to 0.38), 9 of 16 species in Indiana ($\Delta\rho$ range: −0.25 to 0.45), 14 of 15 species in Kansas ($\Delta\rho$ range: −0.09 to 0.37), 14 of 16 species in Michigan ($\Delta\rho$ range: −0.47 to 0.62), 10 of 14 species in Missouri ($\Delta\rho$ range: −0.21 to 0.60), and for all 14 species in Ohio ($\Delta\rho$ range: 0.00–0.68). Likewise, percent overlap of smoothed migration curves generally increased across species and U.S. states with increasing spatial scale. Specifically, overlap of migration curves averaged across U.S. states increased for 12 of 16 species by up to 21% (i.e., *B. canadensis*). Species' migration curves that did not increase in percent overlap from the county or multi-county to U.S. state spatial scale ($\Delta\cap$) were *Bucephala albeola* ($\Delta\cap$ = − 0.01), *Aythya collaris* ($\Delta\cap$ = − 0.02), *C. colombianus* ($\Delta\cap$ = − 0.10), and *O. jamaicensis* ($\Delta\cap$ = − 0.11; Supplementary Material Tables S2 and S3).

At the JV scale, all species' migration curves derived from professional surveys compared to eBird relative abundance were correlated at $\rho \geq 0.70$, except for *Anas rubripes* ($\rho = 0.39$; 95% CI: − 0.04–0.70), *Branta canadensis* ($\rho = 0.45$; 95% CI: 0.03–0.73), and *Aythya americana* ($\rho = 0.69$; 95% CI: 0.38–0.86). Percent overlap was also strong between the 2 migration curves at the JV level where $\cap > 70\%$ for all species except *A. valisineria* ($\cap = 0.35$), *Anas rubripes* ($\cap = 0.42$), *Aythya americana* ($\cap = 0.45$), *A. collaris* ($\cap = 0.49$), and *M. strepera* ($\cap = 0.63$; Figure 3).

Influence of Occurrence and Abundance on Concordance

Correlation between migration curves at the county or multi-county scale depended on a species' observed occurrence and abundance for 12 of 16 species (Figure 4). Specifically, for every 100 additional *Anas rubripes* observed, Spearman's correlation increased by 1.14% (90% CRI: 0.43–2.32%). Similarly, an additional 100 individuals observed during professional surveys increased Spearman's correlation by 4.94% (90% CRI: 1.21–10.05%) for *Aythya americana*, 3.89% (90% CRI: 0.45–7.70%) for *Aix sponsa*, 1.43% (90% CRI: 0.77–2.22%) for *O. jamaicensis*, 1.16% (90% CRI: 0.27–2.19%) for *M. americana*, 0.81% (90% CRI: 0.41–1.26%) for *B. canadensis*, 0.69% (90% CRI: 0.10–1.46%) for *S. discors*, 0.67% (90% CRI: 0.17–1.35%) for *S. clypeata*, 0.32% (90% CRI: 0.07–0.82%) for *Aythya valisineria*, 0.31% (90% CRI: 0.11–0.53%) for *M. strepera*, 0.27% (90% CRI: 0.12–0.44%) for *Anas acuta*, and 0.19% (90% CRI: 0.06–0.34%) for *A. carolinensis* (Supplementary Material Table S1). In contrast, concordance between migration chronologies was not associated with observed maximum abundance of *Bucephala albeola* ($\beta = 1.76e^{-04}$; 90% CRI: $- 2.13e^{-04}$ to $1.10e^{-03}$), *A. platyrhynchos* ($\beta = 3.06e^{-06}$; 90% CRI: $- 7.38e^{-07}$ to $7.22e^{-06}$), *Aythya collaris* ($\beta = 3.58e^{-05}$; 90% CRI: $- 1.02e^{-05}$ to $9.49e^{-05}$), or *C. colombianus* ($\beta = 3.34e^{-03}$; 90% CRI: $- 3.54e^{-03}$ to $1.08e^{-02}$; Figure 4; Supplementary Material Table S1).

DISCUSSION

Reliable data on spatiotemporal waterfowl abundance are required for landscape-scale conservation planning and habitat delivery, yet the extent and frequency of formal surveys traditionally used to estimate these parameters have declined in recent years (Soulliere et al. 2013, Masto et al. 2021). Our work demonstrated that migration chronologies predicted using citizen science data were similar to migration chronologies derived from targeted surveys conducted by wildlife professionals. Concordance between migration chronologies suggested citizen science data could be used to supplement ongoing professional surveys for regional conservation planning during the nonbreeding season across a broad suite of waterfowl species with different life-histories and

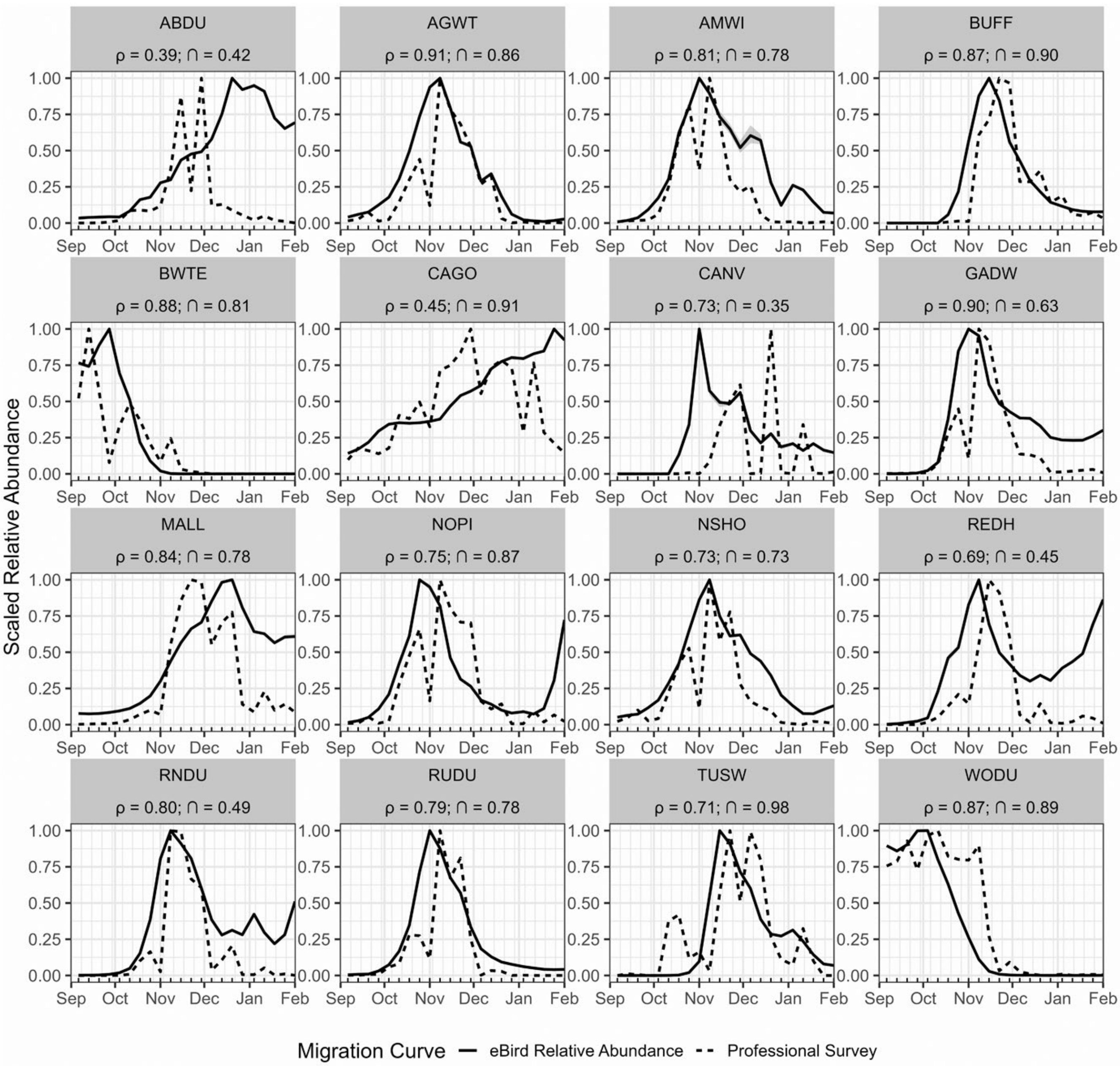

FIGURE 3. Migration curves for the Upper Mississippi / Great Lakes Joint Venture generated from professional surveys (dashed) and by eBird relative abundance (solid). Professional surveys and eBird relative abundance were aligned by week (solid ticks on *x*-axis) to estimate Spearman's rank correlation (ρ) and the percent area of overlap (∩) between the 2 datasets for 16 waterfowl species including *Anas rubripes* (ABDU), *Anas carolinensis* (AGWT), *Mareca americana* (AMWI), *Bucephala albeola* (BUFF), *Spatula discors* (BWTE), *Branta canadensis* (CAGO), *Aythya valisineria* (CANV), *Mareca strepera* (GADW), *Anas platyrhynchos* (MALL), *Anas acuta* (NOPI), *Spatula clypeata* (NSHO), *Aythya americana* (REDH), *Aythya collaris* (RNDU), *Oxyura jamaicensis* (RUDU), *Cynus colombianus* (TUSW), and *Aix sponsa* (WODU). Note the *y*-axis is scaled to the percent of maximum abundance for each dataset for comparisons.

resource requirements (Stuber et al. 2022, Masto et al. 2022). Furthermore, when professional surveys cannot be conducted (e.g., agency closures, budget cuts, inclement weather; Howell et al. 2022) or data are inadequate (Soulliere et al. 2013, 2017; Lancaster et al. 2021), the overlap between migration chronologies detected in our analyses suggest that citizen science data may serve as situational substitutes. Specifically, our analyses support the use of eBird migration chronologies across many species around which the Upper Mississippi / Great Lakes JV conservation planning is based (Soulliere et al. 2017), and likely extend to other regional planning areas (Gulf Coast JV; Lancaster et al. 2021). However, practitioners should be prudent when applying eBird-derived migration chronologies in conservation planning, as they did not align with professional surveys for vagrant species that were not well-distributed across a geography.

Validation Efforts

Although citizen science data generated reliable migration curves for most species, we observed consistently weak concordance at local scales for some species including *Anas rubripes*, *Bucephala albeola*, *Branta canadensis*, and *C.*

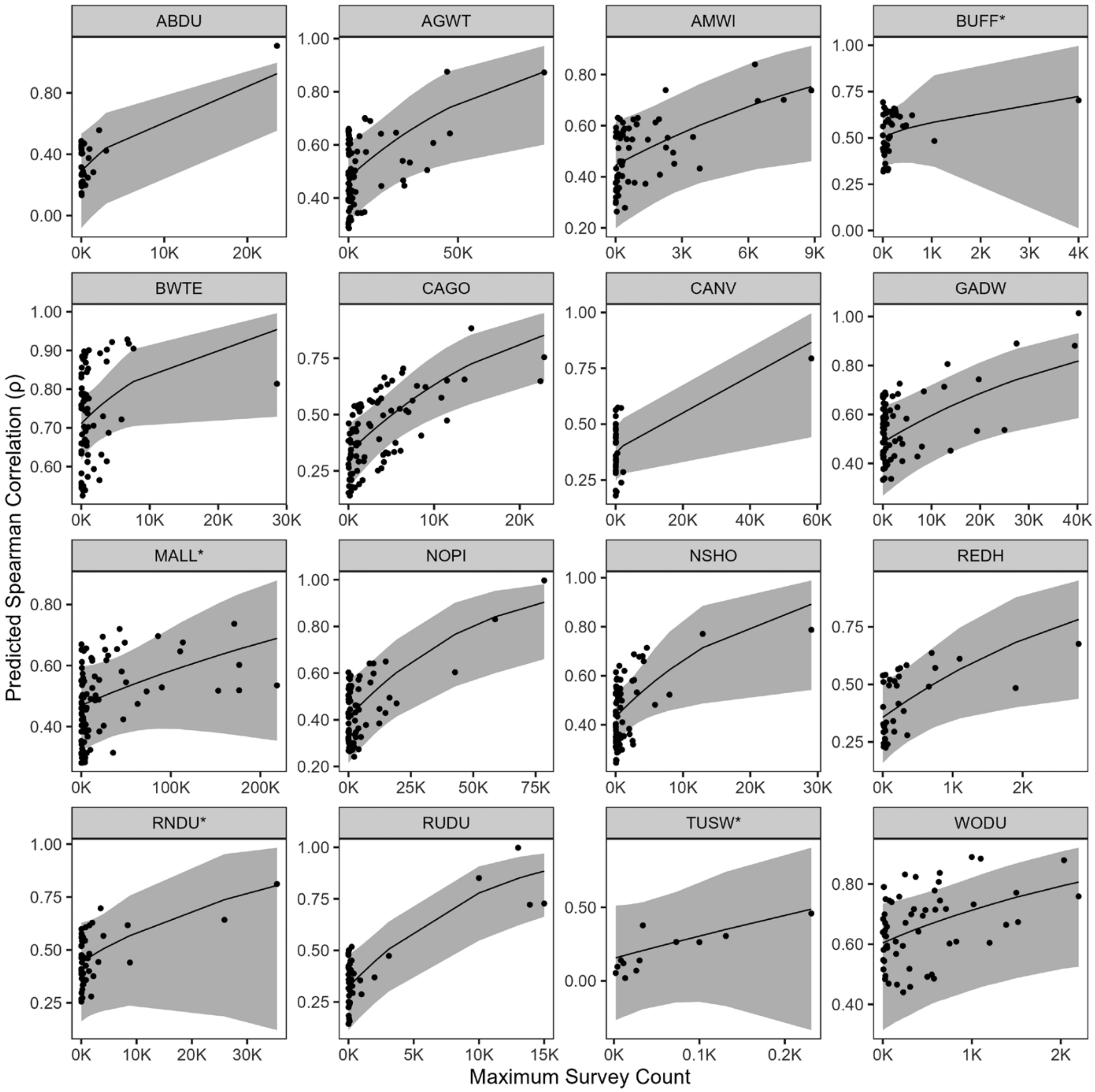

FIGURE 4. Marginal predictions of the correlation between migration chronologies derived from eBird and professional surveys (ρ) for 16 species of waterfowl at the county-multicounty scale. Species included *Anas rubripes* (ABDU), *Anas carolinensis* (AGWT), *Mareca americana* (AMWI), *Bucephala albeola* (BUFF), *Spatula discors* (BWTE), *Branta canadensis* (CAGO), *Aythya valisineria* (CANV), *Mareca strepera* (GADW), *Anas platyrhynchos* (MALL), *Anas acuta* (NOPI), *Spatula clypeata* (NSHO), *Aythya americana* (REDH), *Aythya collaris* (RNDU), *Oxyura jamaicensis* (RUDU), *Cynus colombianus* (TUSW), and *Aix sponsa* (WODU). Asterisks denote the 4 species for which survey count did not influence correlations based on 90% credible intervals overlapping zero. Note the *x*- and *y*-axes differ due to species-specific maximum survey counts and to illustrate positive relationships despite varying correlation coefficients, respectively.

colombianus. Hence, the relationship between concordance of migration curves (ρ) and a species' maximum observed abundance from professional surveys revealed that uncommon (e.g., *A. rubripes*) or cryptic species (e.g., *Aix sponsa*), most often observed at lower densities, had weaker concordance. In contrast, concordance increased as observed abundance increased. In other words, the temporal agreement between eBird relative abundance and professional surveys suffered in areas and for species with few occurrences (i.e., large number of zeros). For example, only 3 of 26 counties in Iowa had non-zero abundance information for *A. rubripes* and only 2 of those counties had eBird relative abundance predictions with which to compare (Supplementary Material Figure S33). Resulting agreement between migration curves at the county level were weak due to so few occurrences of *A. rubripes* in the state (ρ = −0.06 and −0.15). Likewise, 4 *A. rubripes* were observed across 2 counties in Kansas and no eBird relative abundance information was available for comparison in those counties (but eBird data existed in one other county; Supplementary Material Figure S50). Subsequently, when data were pooled to compare migration curves at the U.S. state level, concordance also was predictably weak ($\rho = -0.125$; Supplementary Material Figure S113). Vagrant occurrences in professional surveys or lack of eBird relative abundance information reflects the limited importance of specific regions for a species of interest (e.g., *A. rubripes* in Iowa or Kansas). Such findings are commonplace in citizen science data, wherein predicted relative abundance and population trends were typically less reliable for rare or poorly monitored avian species compared to common (abundant) and well-distributed species (Boersch-Supan et al. 2019). Our analysis highlights this point and emphasizes a need to interpret relationships between eBird relative abundance and professional surveys on a species-by-species basis relative to specific uses of eBird in evidence-based conservation planning and management (Ruiz-Guitierez et al. 2021). In the applied use-case investigated here for nonbreeding waterfowl, migration curves are used in bioenergetic models to estimate required habitat needs based on the population's dietary energy demand within a given geography. Therefore, eBird's utility to generate reliable migration curves for common and abundant species (e.g., *A. platyrhynchos*) is encouraging, given their overwhelming influence on accordant habitat allocation and delivery, compared to less abundant or vagrant species.

Survey effort, observer expertise, and observer consistency for professional surveys could affect their relationships with citizen science data (Masto et al. 2021). Notably, concordance and percent overlap of migration curves were high for most species and area combinations from Illinois and Ohio at the county or multi-county scale, which were also the 2 states that conducted intensive aerial cruise surveys with a consistent expert observer (Forbes Biological Station 2020). Indeed, Stuber et al. (2022) similarly demonstrated a high correlation ($\rho > 0.70$) between eBird relative abundance and Illinois aerial surveys for *M. strepera*, *A. platyrhynchos*, *A. acuta*, and *Aythya collaris*. Conversely, concordance between the 2 data sets was variable for counties or multi-counties in the other 5 U.S. states that conducted vehicle-based ground censuses on public properties (e.g., WMAs or NWRs). Known count and observation biases are associated with both ground and aerial surveys (Elphick 2008, Koneff et al. 2008, Alisauskas and Conn 2019, Gilbert et al. 2021), but aerial surveys cause less disturbance and are more effective at monitoring vast landscapes and habitats less accessible to ground surveyors (e.g., private lands; Masto et al. 2021). Therefore, ground-based censuses of public areas may only represent a fraction of the proportional wetland area within the county compared to eBird's prediction of relative abundance across the entire county, which likely created spatial mismatches that reduced agreement between the 2 datasets. Additionally, temporal mismatches between eBird relative abundance and professional surveys were observed later in the winter planning period (i.e., January–March). In general, the Mississippi Flyway and the Upper Mississippi / Great Lakes JV region have less inundated area available for waterfowl during autumn and early winter, which can congregate ducks and geese locally on our surveyed public areas and riverways that are reliably inundated (Hagy et al. 2014, Masto et al. 2024b). However, as additional floodwater wetlands become available in late winter and early spring, waterfowl may disperse from public and riverine areas to use wetlands that are not covered by ground or aerial surveyors but could still be observed by eBird users (Stuber et al. 2022). Understanding where and why spatial and temporal mismatches occur could be used to modify professional survey protocols or evaluate when eBird is most useful to supplement targeted monitoring programs (Stuber et al. 2022).

The Integrated Waterbird Monitoring and Management (IWMM) by the U.S. Fish and Wildlife Service and the state-operated professional waterfowl surveys provided valuable data that validated eBird-derived migration chronologies at 3 spatial scales relevant to waterfowl conservation planning (Soulliere et al. 2013, Loges et al. 2021). In fact, professionally conducted surveys from the Upper Midwest provided the only contemporary data with adequate temporal frequency and spatial coverage across an entire JV region that we are aware of, with which to compare with eBird relative abundance predictions (Lancaster et al. 2021). Therefore, we recommend that coordinated region-wide monitoring efforts be preserved where practical (Forbes et al. 2020) and resurrected when feasible (Loges et al. 2021), to compare with and supplement novel datasets, such as eBird, as they continue to innovate (Soulliere et al. 2013; Fink et al. 2023a). For example, variable concordance estimates at the county or multi-county scale indicate potential for improved alignment of migration curves at smaller spatial scales. Formal integration of data from the IWMM and other professional surveys with those from citizen science programs may increase the accuracy of abundance estimates and migration chronologies at local scales because citizen scientists often survey areas that professionals do not, and vice versa (Robinson et al. 2018, 2020). We suggest formal integration of multiple datasets, including professional surveys and crowdsourced data, be considered in future research to generate more precise county-level relative abundance estimates. These county-level estimates could then be used to refine the spatial scale at which energy demand is allocated. Nevertheless, our most salient finding was that when citizen science data were compared to professional surveys at spatial scales most relevant to regulatory processes or conservation planning purposes (i.e., U.S. state or JV), concordance increased to reasonable thresholds ($\rho > 0.70$) for abundant and well-distributed species (Aceves-Bueno et al. 2017).

Applications of eBird in Waterfowl Conservation Planning

Overall, validation analyses supported the use of eBird-derived migration curves as supplemental or exclusive data that feed into bioenergetic models to predict energy demand for nonbreeding waterfowl, especially for common and well-distributed species and at state and regional scales. State agencies may also consider using eBird to supplement professional survey data during off-weeks (e.g., Kansas' bi-monthly surveys), adapt survey protocols, or extend their spatial scope of inference beyond survey boundaries to better inform waterfowl season dates (e.g., early *S. discors* hunting seasons) within federal harvest frameworks. When eBird is used to inform restoration, management, or planning objectives at local scales, we recommend practitioners evaluate eBird relative abundance on a species-by-species basis relative to their objectives, use existing knowledge of the ecology of the species (e.g., geographic range) to decide which species to include (Lancaster et al. 2021), or pool predictions across taxonomic guilds to increase occurrence rates for rarer species (Hagy et al. 2022).

Importantly, the eBird Status Product predicts continuous surfaces of relative abundance on a weekly basis across the continental U.S. and Mexico. Therefore, proportional distributions could be derived from eBird relative abundance, providing an alternative framework for setting waterfowl population objectives and estimating dietary energy needs across nonbreeding JV regions for which harvest may not adequately reflect proportional waterfowl distributions. This option could be applied uniformly across JVs, which would be a tremendous advancement towards methodologically consistent stepped-down bioenergetic planning at regional or continental scales. Moreover, a fully integrated eBird stepped-down framework could be adapted to set population objectives for nongame species because harvest records would no longer be required (Vermillion and Lancaster 2022). We view a fully integrated eBird stepped-down framework as an urgent research priority. Until such time, our research suggests that JVs may use eBird-derived migration chronologies within existing objective-setting frameworks for waterfowl conservation planning and delivery in nonbreeding regions.

Supplementary material

Supplementary material is available at *Ornithological Applications* online.

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Ethics statement

Everyone intimately involved in this study is included in the by-line. All authors agreed to the final published version of the manuscript and do not declare conflicts of interest.

Conflict of interest statement

The authors declare no conflicts of interest.

Author contributions

N.M.M., K.D.D., O.J.R., M.G.B., and K.M.R. conceived project ideas and development. N.M.M., J.M.O., B.A., T.B., M.E., O.E.J., B.W.L., D.M., W.A.P., and R.V. collected or compiled data. N.M.M., K.D.D., and O.J.R. performed analyses, interpreted results, and produced visualizations. NMM wrote the first draft and all authors contributed critically to final drafts. M.G.B. and O.J.R. administered the project and secured funding. Everyone intimately involved in this study is included in the by-line. All authors agreed to the final published version of the manuscript.

Data Availability

Analyses reported in this article can be reproduced using the data and code provided by Masto et al. (2025). eBird relative abundance can be accessed using the ebirdst package in R.

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